SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.5: Rates of Reactions — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.5: Rates of Reactions
Driving Question	DRIVING QUESTION: Why do some chemical reactions happen in a flash while others take days — and how can we control the speed of reactions to make life in Kenya safer, healthier, and more productive?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Is a Rate of Reaction? — Anchoring Phenomenon Launch**	Students observed two contrasting phenomena to anchor the unit: (1) a fast reaction — effervescent antacid tablets (or sodium thiosulfate + hydrochloric acid producing a sulfur precipitate that quickly obscures a drawn cross) fizzing vigorously within seconds; and (2) a slow reaction — photographs/samples of a rusting jembe (hoe) or browning of a sliced avocado from a Kenyan market, taking hours to days. Teacher should display both side-by-side and ask: 'What is different about how quickly these two changes happen?' Students also observed that the sodium thiosulfate–HCl reaction produces a cloudy yellow precipitate at a visible, measurable rate — the anchoring experiment for the whole unit.	Students learned that chemical reactions do not all happen at the same speed — some are almost instantaneous (explosion of a gas stove burner being lit in a Nairobi kitchen) while others are extremely slow (iron rusting on a Mombasa fishing boat over months). They learned that the term 'rate of reaction' refers to how fast or slow a chemical reaction proceeds, and that this speed can be observed by watching reactants disappear or products appear. Students also began building the unit's Driving Question Board (DQB), posting initial 'What we notice / What we wonder' sticky notes.	Pedagogical note: This lesson is intentionally observational and conceptual — resist the urge to define rate mathematically yet (that is Lesson 2). The goal is productive confusion: students should leave asking 'WHY are some reactions faster?' which sets up the entire unit. Cold-call at least 4 students to describe what they noticed. Use wait time of 8–10 seconds before accepting answers. Anchor the phenomenon by naming it explicitly: 'For the next 8 lessons, everything we learn will help us explain why that cross disappears at different speeds under different conditions — and how that connects to rusting jembes, cooking ugali, and making medicine safer in Kenya.' The sulfur precipitate forming in the $Na_2S_2O_3$ + HCl reaction is the shared class phenomenon: $S_2O_3^{2-}(aq) + 2HCl(aq) \rightarrow S(s) + SO_2(g) + H_2O(l) + 2Cl^-(aq)$. The solid sulfur (S) makes the solution cloudy.
Lesson 2 **What Is Rate of Reaction? — Measuring**	Students repeated the sodium thiosulfate + hydrochloric acid cross-disappearance experiment under controlled conditions, this	Students learned the formal definition: Rate of reaction = change in measurable quantity ÷ time taken. For the cross-disappearance	Pedagogical note: Emphasise the inverse relationship — 1/t is the rate, so if Group A's cross disappeared in 20 s and Group B's in

the Speed of Change	time timing (in seconds) how long it takes for the cross drawn beneath the conical flask to disappear from view as sulfur precipitate builds up. One standard concentration and temperature was used. Multiple groups recorded their times and compared results. Students also examined a rate–time graph drawn on the board showing how the concentration of a reactant decreases over time (a curve that starts steep and flattens), and a product–time graph that is the mirror image.	experiment, rate is expressed as 1/t (units: s^{-1}), where t is the time in seconds for the cross to disappear — a longer time means a slower rate, and a shorter time means a faster rate. Students learned that rate can be measured by: (a) change in mass (e.g., CO_2 gas escaping from marble chips + HCl, measured on a balance — relevant to antacid tablet quality control), (b) change in volume of gas produced, (c) change in colour/turbidity (the cross method), or (d) change in pH. Students also learned that reaction rate is highest at the start (most reactant particles present) and decreases as reactants are used up.	40 s, Group A's rate (1/20 = 0.05 s^{-1}) is TWICE Group B's rate (1/40 = 0.025 s^{-1}). Many students confuse 'time taken' with 'rate' — explicitly address this misconception. Use a real Kenyan context: 'A Kericho tea factory needs to know how fast a fermentation reaction proceeds — they measure rate, not just time, so they can compare batches.' Connect back to Lesson 1 anchoring phenomenon: the cross-disappearance time IS our operational measure of rate for this unit. Update the DQB: 'We can now measure rate. But what CHANGES rate?' to preview Lessons 3–7.
Lesson 3 **Concentration and Collision Frequency: Why Does a More Concentrated Acid React Faster?**	Students carried out the sodium thiosulfate + HCl cross-disappearance experiment at four different concentrations of sodium thiosulfate (e.g., 40, 30, 20, 10 cm^3 of $Na_2S_2O_3$ diluted to the same total volume with water each time), keeping HCl volume and temperature constant. They recorded the time (t) for the cross to disappear at each concentration, calculated 1/t for each, and plotted a graph of 1/t (rate, y-axis) against concentration of $Na_2S_2O_3$ (x-axis). The graph produced a straight line through the origin, showing a directly proportional relationship.	Students learned that increasing the concentration of a reactant increases the rate of reaction. The explanation lies in collision theory: a higher concentration means more solute particles are dissolved in the same volume of solution, so particles are closer together and collide more frequently. More frequent collisions mean more successful collisions per second (those with energy ≥ activation energy), so more product is formed per second — the rate increases. Conversely, diluting a reactant (as happens naturally as a reaction proceeds) reduces collision frequency and the rate slows. Real-life Kenya connections established: higher concentration of acid in a car battery causes faster corrosion of terminals; a more concentrated bleach solution disinfects a Nairobi hospital surface faster.	Pedagogical note: Collision theory is the explanatory framework for ALL factors in this sub-strand — establish it firmly here so it can be referenced in Lessons 4, 5, 6, and 7. Draw a particle diagram on the board: low concentration (few particles, far apart, rare collisions) vs. high concentration (many particles, close together, frequent collisions). Cold-call 3 students to explain in their own words why rate increases. Anticipated misconception: students may say 'more concentration means more energy' — clarify that concentration affects FREQUENCY of collisions, not the energy of individual collisions (that is temperature's role, coming in Lesson 4). Update the DQB with the answer to the concentration question and post the new wonder: 'Does heating also increase collision frequency?'
Lesson 4 **Temperature and Activation Energy: Why Does Heating Speed Up Reactions?**	Students repeated the cross-disappearance experiment at four different temperatures (e.g., ~15°C, 25°C, 35°C, 45°C) using ice-water baths and warm water baths, keeping concentration of both reactants constant. They recorded t and calculated 1/t at each temperature, then plotted rate (1/t) vs. temperature (°C). The graph showed a curve (exponential increase), not a straight line — a key observation distinguishing temperature from concentration. Students	Students learned that increasing temperature increases the rate of reaction for two reasons: (1) particles move faster (have greater kinetic energy), so they collide MORE FREQUENTLY; and (2) more particles have energy greater than or equal to the activation energy (E_a) — the minimum energy needed for a successful collision — so a greater PROPORTION of collisions are successful. The combined effect is a large increase in rate for a	Pedagogical note: The activation energy concept is new and abstract — use an analogy: 'Think of activation energy like the hill you must push a cart over before it rolls downhill on its own. At low temperature, few particles have enough energy to get over the hill. At high temperature, many more can.' Draw a Maxwell–Boltzmann-style energy distribution curve (simplified) showing the shift at higher temperature. Emphasise that temperature affects BOTH

	also observed (via teacher demonstration or video) that food kept in a Nairobi market at room temperature spoils much faster than food kept in a refrigerator — connecting rate of biological/chemical reactions to temperature in daily Kenyan life.	relatively small rise in temperature (roughly doubling for every 10°C rise, as a rule of thumb). Students learned the definition of activation energy. They connected this to: preserving fish at Lake Victoria using ice to slow spoilage reactions; refrigerating sukuma wiki to slow wilting/oxidation; and why a Rift Valley farmer's crop stored in a cool granary lasts longer.	frequency AND proportion of successful collisions, unlike concentration which affects frequency only. This is why the rate–temperature graph curves upward rather than being linear. Update DQB: mark temperature as explained and post: 'What about the SIZE of solid particles — does that affect rate?'
Lesson 5 **Surface Area and Reaction Rate — Grinding, Crushing, and the Secret Inside Solid Reactants**	Students compared the reaction of large marble chips (calcium carbonate) vs. powdered marble with excess dilute hydrochloric acid, measuring the volume of CO_2 gas produced over time (or mass loss on a balance). Both experiments used the same mass of marble and the same volume and concentration of HCl. Students observed that the powdered marble produced gas much faster (steeper initial gradient on the volume–time graph) but both reactions produced the same total volume of gas (same mass of reactant used). A second demonstration used a whole dried maize grain vs. maize flour ignited — the flour flashed dramatically (teacher-led, safety-critical), connecting to grain-dust explosions in flour mills.	Students learned that increasing the surface area of a solid reactant increases the rate of reaction. When a solid is broken into smaller pieces (or ground to powder), the total surface area exposed to the other reactant increases enormously, while the total mass (and therefore the total number of moles of reactant) remains the same. More surface area means more solid particles are in direct contact with the solution/gas reactant, leading to more frequent collisions at the solid–liquid (or solid–gas) interface. Crucially, students learned that surface area affects RATE but NOT the total amount of product formed (the yield), which depends only on the limiting reactant. Kenya contexts: grinding maize into unga (flour) makes it cook faster; powdered charcoal catches fire more easily than a charcoal lump; finely divided metal catalysts in industrial processes (like fertiliser manufacture) react faster.	Pedagogical note: The 'same total gas, different rate' result is a powerful conceptual anchor — both curves on the volume–time graph must reach the same plateau. If student data shows different plateaus, troubleshoot: likely a different mass of marble was used. The maize flour demonstration MUST be done by the teacher only, outdoors or in a well-ventilated area, with a small quantity — it illustrates grain-dust explosion hazards relevant to Kenyan posho mills. This is an excellent PCIs (Safety) integration moment. Cold-call students: 'If we used the same mass of powdered marble but HALF the volume of HCl, what would change?' (rate stays high initially, but total gas produced halves — HCl is now limiting). Update DQB: add surface area as explained, post: 'Can we speed up reactions without changing concentration, temperature, or particle size?'
Lesson 6 **Pressure and Reaction Rate — When Gases Are the Reactants**	Students examined data sets and graphs (printed or projected) showing the effect of pressure on gas-phase reactions, since direct pressure experiments are difficult to perform safely at school level. Teacher demonstrated (or showed video of) a syringe containing a small amount of gas being compressed, visually showing particles becoming more crowded. Students also analysed a case study: the Haber process for making ammonia ($N_2 + 3H_2 \rightleftharpoons 2NH_3$), which is used to produce nitrogenous fertilisers critical to Kenyan agriculture in the Rift Valley and Central regions, is carried out at high pressure	Students learned that for reactions involving gases, increasing the pressure increases the rate of reaction. Increasing pressure on a gas decreases the volume it occupies, so the same number of gas particles are packed into a smaller space — this is equivalent to increasing the concentration of the gas. Higher particle density means more frequent collisions, and therefore a faster rate. This is directly analogous to concentration for solutions. Students applied this to: the Haber process (high pressure → faster NH_3 production → more fertiliser for Kenyan farms); LPG cooking gas in a pressurised cylinder reacting faster	Pedagogical note: Pressure is conceptually the 'concentration equivalent for gases' — explicitly make this link to build coherence. Students often forget that pressure only applies to gases; pre-empt this by asking: 'Would squeezing a bottle of sodium thiosulfate solution change the rate? Why not?' (Liquids do not compress significantly, so particle density does not change.) The Haber process case study is an excellent opportunity to connect chemistry to Kenyan food security — urea and CAN (calcium ammonium nitrate) fertilisers used by Kenyan smallholder farmers are downstream products of ammonia

	(150–300 atm) specifically to increase reaction rate.	when the valve is fully open; and combustion reactions in a car engine's compressed cylinder. Students also noted that pressure has negligible effect on reactions in solution (liquids are nearly incompressible).	synthesis. Update DQB: add pressure as explained. Post the final factor wonder: 'Is there a substance that can speed up a reaction without being used up itself?'
Lesson 7 **Light and Catalysts — The Final Rate Factors**	Students observed two demonstrations: (1) Catalyst demo — hydrogen peroxide (H_2O_2) decomposing slowly at room temperature vs. decomposing rapidly (oxygen gas bubbling vigorously) when a small amount of manganese(IV) oxide (MnO_2) powder is added. After the reaction, students filtered and dried the MnO_2 and confirmed it was recovered unchanged — proving it was not consumed. (2) Light demo — teacher showed photographic paper or a silver salt solution (AgCl) darkening rapidly when exposed to light vs. remaining pale in a dark cupboard, illustrating that light energy can initiate or accelerate photochemical reactions (relevant to photography and to vitamin D synthesis in human skin under Kenyan sunlight).	Students learned that: (a) A catalyst is a substance that increases the rate of a reaction by providing an alternative reaction pathway with a lower activation energy, without itself being permanently consumed. Because the new pathway has a lower E_a, a greater proportion of collisions have sufficient energy to react, so rate increases. Catalysts are vital in industry (MnO_2 in H_2O_2 decomposition; iron in the Haber process; enzymes as biological catalysts in fermentation of uji or busaa) and in the body (enzymes digesting ugali starch). (b) Light (photons) can supply the activation energy needed to start some reactions — these are photochemical reactions. Examples: photosynthesis in Kenyan maize crops; bleaching of coloured fabrics left in Nairobi sunshine; silver halide darkening in old-style photography.	Pedagogical note: The catalyst demonstration is highly engaging — ensure H_2O_2 used is the standard lab concentration (20 vol / ~6%) for safety. Emphasise the key catalyst property: it is RECOVERED UNCHANGED at the end. This is why industrial catalysts are economically valuable — they do not need to be replaced after every reaction cycle. Link to enzymes explicitly: 'The amylase enzyme in your saliva is a biological catalyst — it speeds up the breakdown of ugali starch in your mouth without being consumed.' For light: clarify that light is an energy source that can replace thermal activation energy in photochemical reactions, not a substance. Update the DQB: all five factors (concentration, temperature, surface area, pressure, catalyst/light) should now be marked as explained. Begin the final model-building synthesis.
Lesson 8 **Controlling Reaction Rates — Synthesis, Application, and Solutions for Kenya**	Students reviewed their cumulative collision-theory model (built across Lessons 3–7) and applied it to three real Kenyan scenarios presented as case studies: (1) A posho mill operator in Eldoret wants to prevent a grain-dust explosion — which rate factors are relevant and how should she control them? (2) A pharmacist in Kisumu needs to store liquid medicine (which decomposes via a chemical reaction) so that it remains effective for 18 months — which factors should be controlled and how? (3) A Kenyan petroleum refinery (Mombasa) uses catalytic cracking to break large hydrocarbon molecules into useful fuels — explain the role of the catalyst and temperature in this industrial process. Students worked in groups and presented their analyses.	Students synthesised all five rate factors — concentration, temperature, surface area, pressure, and catalysts/light — and applied them to real-world Kenyan contexts. They learned that controlling reaction rates is not about making all reactions as fast as possible; sometimes slowing a reaction is the goal (food preservation, medicine storage, preventing explosions). They demonstrated understanding that collision theory provides a single, unified explanation for all five factors: any change that increases the frequency of collisions OR increases the proportion of collisions that have energy ≥ activation energy will increase the rate. Students completed their final cumulative model and revisited the driving question, constructing written or diagrammatic answers explaining why	Pedagogical note: This lesson is the unit capstone — prioritise student voice and application over new content delivery. The three case studies should be genuinely open-ended; accept any well-reasoned answer grounded in collision theory. For the posho mill scenario, key answers include: reduce surface area of dust exposure (engineering controls), avoid ignition sources (remove activation energy source), and ensure ventilation (reduce concentration of dust in air). For the medicine scenario: refrigerate (lower temperature slows decomposition rate), use airtight containers (prevent concentration increase of reactive gases like O_2), store away from light (prevent photochemical decomposition). Conduct a whole-class DQB review: read the original driving

		some reactions 'happen in a flash' and how Kenyan industries, farmers, and health workers can control reaction rates for safety, health, and productivity.	question aloud, then ask students to vote on whether they can now answer it fully. Celebrate the journey from 'I wonder why some reactions are fast' to a complete collision-theory model. Assign the end-of-sub-strand assessment task.